Bipolar Junction Transistors (BJTs)

Electronic Devices and Circuits

Comprehensive Study Report

Based on Course Lecture Material

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1 Introduction to Bipolar Junction Transistors

1.1 Historical Background and Evolution

The transistor, a fundamental building block of modern electronics, was invented at Bell Laboratories in 1947 by John Bardeen, Walter Brattain, and William Shockley. Prior to the invention of the transistor, electronic systems relied on vacuum tubes. Vacuum tubes were physically large, operated at high temperatures, and consumed significant amounts of power, with a considerable portion of that power wasted as heat. The advent of the solid-state transistor revolutionized electronics by addressing these massive inefficiencies.

By the 1970s and 1980s, the Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET) became familiar, and by 2014, it became the most widely used electronic device, making CMOS the technology of choice for integrated circuits. However, the Bipolar Junction Transistor (BJT) remains the preferred device in demanding analog and digital applications, particularly in very-high-frequency and high-speed circuits. Furthermore, bipolar transistors are often combined with MOSFETs to create BiCMOS technology, merging the high-input-impedance and low-power benefits of MOSFETs with the high-current-driving capability of BJTs.

1.2 Transistor Structure and Function

The term “bipolar” signifies that charge carriers of both polarities—electrons and holes—participate in the current-conduction process. A BJT consists of three semiconductor regions: the Emitter (E), the Base (B), and the Collector (C).

Depending on the arrangement of the doped semiconductor layers, BJTs are classified into two main types:

- **NPN Transistor:** Comprises a p-type base sandwiched between an n-type emitter and an n-type collector.
- **PNP Transistor:** Comprises an n-type base sandwiched between a p-type emitter and a p-type collector.

The emitter region is heavily doped to ensure a high concentration of majority carriers, while the base region is lightly doped and physically very thin. The collector region is moderately doped. In a physical integrated circuit, the structure is not perfectly symmetrical; the collector virtually surrounds the emitter region to efficiently capture injected carriers. Therefore, the emitter and collector cannot be simply interchanged.

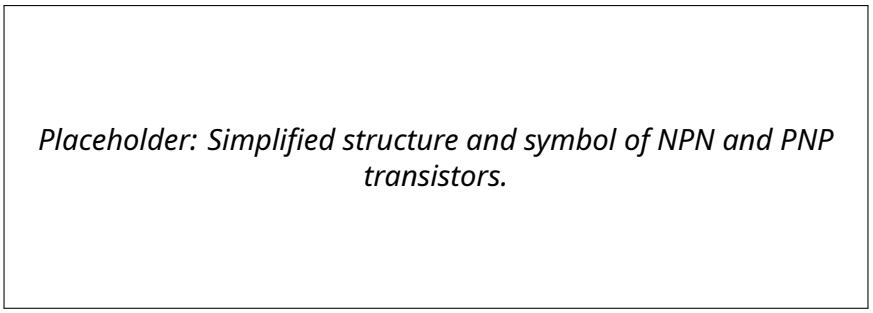


Figure 1: Structure and circuit symbols for NPN and PNP transistors.

2 Device Structure and Physical Operation

2.1 Modes of Operation

A BJT contains two internal pn junctions: the Emitter-Base Junction (EBJ) and the Collector-Base Junction (CBJ). The bias state (forward or reverse) of these two junctions dictates the operating mode of the transistor, as summarized in Table 1.

Mode	**Emitter-Base Junction (EBJ)**	**Collector-Base Junction (CBJ)**
Active	Forward	Reverse
Saturation	Forward	Forward
Cut-off	Reverse	Reverse

Table 1: BJT Operating Modes and Bias Conditions.

The **active mode** is utilized when the transistor is operated as an amplifier. Conversely, both the **cut-off** and **saturation modes** are used as extreme states for switching applications, such as in logic circuits.

2.2 Operation in the Forward-Active Mode

In an NPN transistor biased in the active mode, the EBJ is forward-biased and the CBJ is reverse-biased. Because the emitter is heavily doped (n-type) relative to the base (p-type), a vast number of electrons are injected from the emitter into the base.

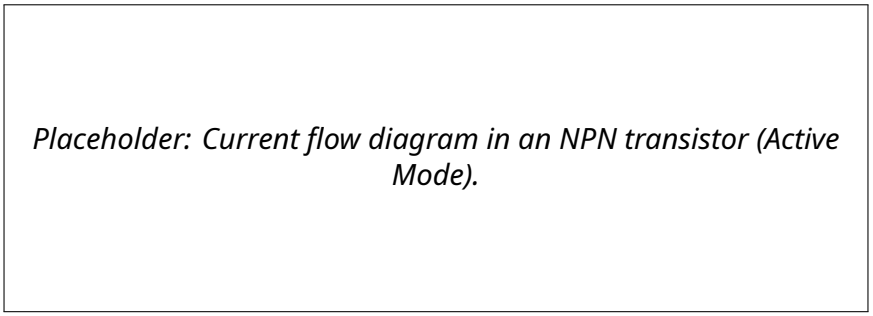


Figure 2: Current flow in an NPN transistor biased to operate in the active mode.

2.2.1 Minority-Carrier Distribution

In the narrow base region, the injected electrons act as minority carriers. Their concentration at the EBJ boundary, $n_p(0)$, is proportional to the exponential of the applied base-emitter voltage:

$$n_p(0) = n_{p0} e^{v_{BE}/V_T} \tag{1}$$

where n_{p0} is the thermal-equilibrium minority-carrier concentration in the base. Because the base width W is extremely small, the excess electron concentration decays almost linearly across the base. The electron diffusion current I_n is proportional to the slope of this profile:

$$I_n = A_E q D_n \frac{dn_p(x)}{dx} = A_E q D_n \left(-\frac{n_p(0)}{W} \right) \tag{2}$$

where A_E is the cross-sectional area of the EBJ, q is the elementary charge, and D_n is the electron diffusivity.

2.2.2 Terminal Currents: Collector, Base, and Emitter

Collector Current (i_C): The electrons that successfully traverse the base without recombining are swept across the reverse-biased CBJ into the collector. The collector current is thus given by:

$$i_C = I_S e^{v_{BE}/V_T} \tag{3}$$

where the saturation current I_S is:

$$I_S = \frac{A_E q D_n n_i^2}{N_A W} \tag{4}$$

Base Current (i_B): The base current consists of two components: 1. i_{B1}: Holes injected from the base into the emitter (proportional to e^{v_{BE}/V_T}). 2. i_{B2}: Holes supplied by the external circuit to replace those lost to recombination with electrons in the base (proportional to the number of injected electrons).

The total base current can be expressed as a fraction of the collector current:

$$i_B = \frac{i_C}{\beta} = \left(\frac{I_S}{\beta} \right) e^{v_{BE}/V_T} \tag{5}$$

where β is the **common-emitter current gain**. To achieve a high β, the base must be thin and lightly doped compared to the emitter. Typical β values range from 50 to 200.

Emitter Current (i_E): By Kirchhoff's Current Law, the emitter current is the sum of the collector and base currents:

$$i_E = i_C + i_B = \left(\frac{\beta + 1}{\beta} \right) i_C \tag{6}$$

Alternatively, we define the **common-base current gain** (α) such that $i_C = \alpha i_E$. The relationship between α and β is:

$$\alpha = \frac{\beta}{\beta + 1} \quad \text{and} \quad \beta = \frac{\alpha}{1 - \alpha} \tag{7}$$

2.3 Large-Signal Equivalent Circuit Models

The operation of the BJT can be modeled using diodes and dependent current sources. A popular model represents the collector current as a voltage-controlled current source dependent on v_{BE}. Because $i_C = \beta i_B$, the model can also be drawn as a current-controlled current source.

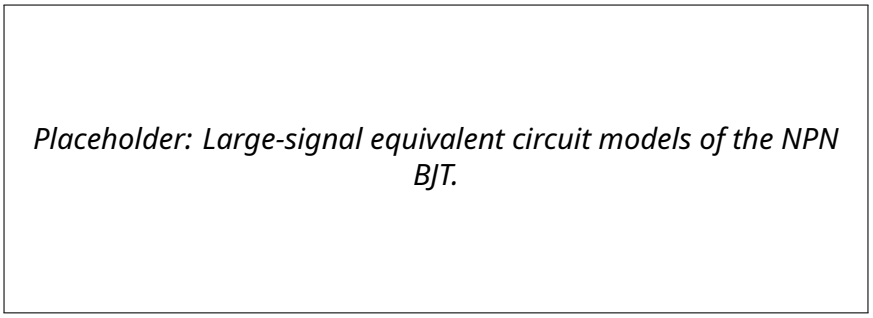


Figure 3: Large-signal equivalent circuit models for an NPN BJT in active mode.

3 BJT Current–Voltage Characteristics

3.1 Active vs. Saturation Modes

An NPN transistor with a forward-biased EBJ ($V_{BE} \approx 0.7$ V) operates in the active mode provided the collector voltage does not fall below the base voltage by more than approximately 0.4 V. If $v_{CB} < -0.4$ V, the CBJ becomes forward-biased, and the transistor enters the saturation region. In saturation, the collector can no longer effectively sweep out injected carriers, and the collector current rapidly falls.

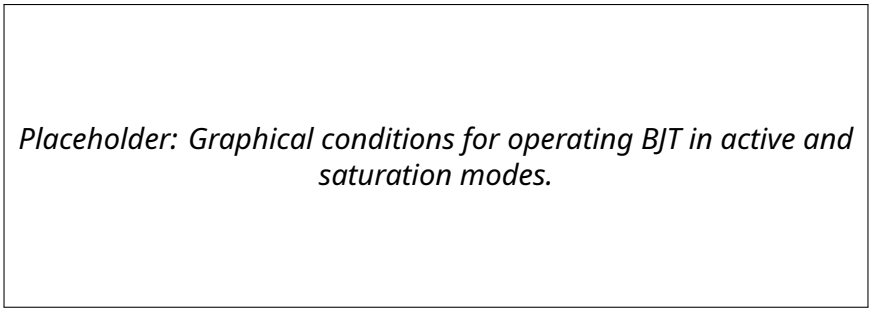


Figure 4: Voltage conditions distinguishing active and saturation modes.

3.2 Temperature Dependence and Reverse Current

The reverse current across the collector-base junction is denoted I_{CBO} (emitter open-circuited). Like all semiconductor leakage currents, I_{CBO} is highly temperature-dependent, approximately doubling for every 10°C rise in temperature.

Additionally, the required base-emitter voltage V_{BE} for a given collector current decreases

by approximately 2 mV/°C as temperature rises. For example, if $I_C = I_S e^{V_{BE}/V_T}$, at a constant I_C, the $i_C - v_{BE}$ curve shifts to the left at higher temperatures.

3.3 The Early Effect and Base-Width Modulation

In the active region, increasing the collector-emitter voltage v_{CE} increases the reverse-bias on the CBJ. This widens the depletion region into the base, thereby decreasing the effective base width W. Because I_S is inversely proportional to W, i_C increases slightly with v_{CE}. This phenomenon is known as the **Early Effect**.

Graphically, extrapolating the linear portions of the active-region output characteristics (i_C vs. v_{CE}) reveals that they all intersect the negative voltage axis at a single point, denoted $-V_A$. V_A is the Early voltage, typically between 10 V and 100 V.

The modified collector current equation is:

$$i_C = I_S e^{v_{BE}/V_T} \left(1 + \frac{v_{CE}}{V_A}\right) \tag{8}$$

This slope in the output characteristics implies a finite output resistance, r_o, given by:

$$r_o = \frac{V_A}{I'_C} \approx \frac{V_A}{I_C} \tag{9}$$

3.4 Saturated Transistor Equivalent Circuit

When driven deeply into saturation, the incremental β drops significantly below the active-region β. The transistor can be modeled as a small voltage drop between the collector and emitter. For a saturated BJT, the saturation voltage is typically approximated as:

$$|V_{CE\text{sat}}| \approx 0.2 \text{ V} \tag{10}$$

4 DC Analysis of BJT Circuits

Performing a DC analysis of a BJT circuit entails determining all node voltages and branch currents. The standard assumption for active-mode operation is a base-emitter voltage drop of $V_{BE} \approx 0.7$ V.

4.1 Worked Example 1

Consider an NPN transistor circuit with the following parameters: a base bias voltage $V_B = +4$ V, an emitter resistor $R_E = 3.3$ kΩ connected to ground, a collector resistor $R_C = 4.7$ kΩ tied to a $V_{CC} = +10$ V supply, and $\beta = 100$.

Solution Sequence: 1. **Determine Emitter Voltage (V_E):**

$$V_E = V_B - V_{BE} = 4.0 \text{ V} - 0.7 \text{ V} = 3.3 \text{ V}$$

2. **Calculate Emitter Current (I_E):**

$$I_E = \frac{V_E}{R_E} = \frac{3.3\ \text{V}}{3.3\ \text{k}\Omega} = 1.0\ \text{mA}$$

3. **Calculate Collector Current (I_C):** Since $\alpha = \frac{\beta}{\beta+1} = \frac{100}{101} \approx 0.99$:

$$I_C = \alpha I_E = 0.99 \times 1.0\ \text{mA} = 0.99\ \text{mA}$$

4. **Determine Collector Voltage (V_C):**

$$V_C = V_{CC} - I_C R_C = 10\ \text{V} - (0.99\ \text{mA})(4.7\ \text{k}\Omega) = 10\ \text{V} - 4.653\ \text{V} = 5.347\ \text{V}$$

5. **Verify Active Mode:** $V_C(5.347\ \text{V}) > V_B(4.0\ \text{V})$, indicating the CBJ is indeed reverse-biased, confirming active mode operation.

5 Small-Signal Operation and Models

For an amplifier, a small time-varying signal is superimposed on the DC bias. The total base-emitter voltage is $v_{BE} = V_{BE} + v_{be}$.

5.1 Transconductance (g_m)

Applying the exponential relationship, the total collector current is:

$$i_C = I_S e^{(V_{BE}+v_{be})/V_T} = I_C e^{v_{be}/V_T} \tag{11}$$

If the signal is small ($v_{be} \ll V_T$, practically < 10 mV), we linearize using a Taylor series approximation $e^x \approx 1 + x$:

$$i_C \approx I_C \left(1 + \frac{v_{be}}{V_T}\right) = I_C + \frac{I_C}{V_T} v_{be} \tag{12}$$

The signal component of the collector current is $i_c = g_m v_{be}$, where g_m is the transconductance:

$$g_m = \frac{I_C}{V_T} \tag{13}$$

Transconductance is directly proportional to the DC bias current I_C.

5.2 Input Resistance

Looking into the Base (r_π): The base signal current is $i_b = i_c/\beta$. The small-signal input resistance at the base is:

$$r_\pi \equiv \frac{v_{be}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B} \tag{14}$$

Looking into the Emitter (r_e): The emitter signal current is $i_e = i_c/\alpha$. The small-signal input resistance at the emitter is:

$$r_e \equiv \frac{v_{be}}{i_e} = \frac{V_T}{I_E} = \frac{\alpha}{g_m} \approx \frac{1}{g_m} \tag{15}$$

These two resistances are related by $r_\pi = (\beta + 1) r_e$.

5.3 Small-Signal Equivalent Circuits

To perform AC analysis, all DC voltage sources are short-circuited to ground, and DC current sources are opened. The BJT is replaced with its small-signal model.

Hybrid-π Model: Uses r_π at the input and a dependent current source ($g_m v_{be}$ or βi_b). The output resistance r_o can be placed in parallel with the dependent source. **T-Model:** Uses r_e connected to the emitter node and a dependent source (αi_e or $g_m v_{be}$) in the collector branch.

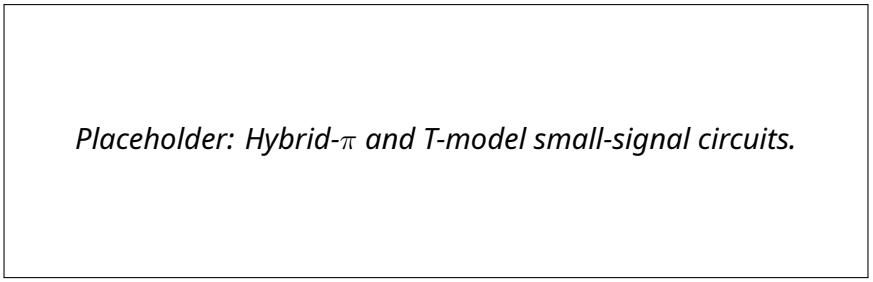


Figure 5: The Hybrid-π and T-model small-signal equivalent circuits.

6 Basic Amplifier Configurations

Amplifiers are functionally characterized by three primary metrics: 1. **Input Resistance (R_{in}):** Represents the loading effect of the amplifier on the signal source. 2. **Open-Circuit Voltage Gain (A_{vo}):** The intrinsic voltage gain when the load resistance $R_L \to \infty$. 3. **Output Resistance (R_o):** The equivalent resistance looking back into the amplifier output with v_{sig} set to zero.

The overall voltage gain G_v, factoring in the source resistance R_{sig} and load R_L, is derived using voltage dividers:

$$v_i = v_{sig}\left(\frac{R_{in}}{R_{in}+R_{sig}}\right) \tag{16}$$

$$v_o = (A_{vo}v_i)\left(\frac{R_L}{R_L+R_o}\right) \tag{17}$$

$$G_v \equiv \frac{v_o}{v_{sig}} = \left(\frac{R_{in}}{R_{in}+R_{sig}}\right) A_{vo} \left(\frac{R_L}{R_L+R_o}\right) \tag{18}$$

7 Common-Emitter Amplifiers

The Common-Emitter (CE) configuration is the most widely used BJT amplifier.

7.1 Basic CE Amplifier (Without Emitter Resistance)

Using the Hybrid-π model for a standard CE amplifier (with collector resistor R_C):

- **Input Resistance:** $R_{in} = r_\pi$
- **Voltage Gain:** $A_{vo} = -g_m R_C$
- **Output Resistance:** $R_o = R_C$ (assuming $r_o \to \infty$)

The negative sign in the gain equation indicates a 180° phase shift (inversion).

7.2 CE Amplifier with Emitter Resistance (R_e)

Inserting a resistance R_e in the emitter branch provides negative feedback, known as emitter degeneration. Using the T-model is highly effective here.

- **Input Resistance:** The resistance looking into the base is the emitter resistance multiplied by $(\beta + 1)$, known as the resistance-reflection rule:

$$R_{in} = (\beta + 1)(r_e + R_e) \tag{19}$$

- **Voltage Gain:**

$$A_{vo} = -\frac{\alpha R_C}{r_e + R_e} \approx -\frac{g_m R_C}{1 + g_m R_e} \tag{20}$$

Implications of Emitter Degeneration: 1. The input resistance R_{in} is increased by a factor of $(1 + g_m R_e)$. 2. The voltage gain A_{vo} is reduced by the same factor $(1 + g_m R_e)$. 3. The circuit is capable of handling larger input signals without severe nonlinear distortion. 4. The overall voltage gain becomes far less dependent on the unstable β parameter.

8 Transistor Breakdown and Temperature Effects

8.1 Voltage Breakdown Regions

Like pn junction diodes, BJTs suffer from avalanche breakdown at high reverse-bias voltages.

- BV_{CBO} **(Common-Base):** Breakdown voltage of the CBJ with the emitter open. Typically > 50 V.
- BV_{CEO} **(Common-Emitter):** Breakdown voltage from collector to emitter with the base open. BV_{CEO} is generally about half the value of BV_{CBO}. This is sometimes referred to as the sustaining voltage.

8.2 Beta and Temperature Dependencies

The current gain β is not a static constant. It varies heavily with both I_C and temperature.

- **Current Dependence:** β is small at very low I_C, reaches a maximum at moderate operating currents, and then drops again at very high I_C.

- **Temperature Effects:** β increases with temperature. Furthermore, an increase in β causes I_C to increase. The increased current generates more localized heat, potentially triggering thermal runaway. This necessitates robust DC biasing schemes to stabilize the operating point against temperature fluctuations.

Formula Sheet: BJT Key Equations

DC Currents and Current Gains

$$i_C = I_S e^{v_{BE}/V_T} \tag{21}$$
$$i_E = i_C + i_B \tag{22}$$
$$i_C = \alpha i_E \tag{23}$$
$$i_C = \beta i_B \tag{24}$$
$$\alpha = \frac{\beta}{\beta + 1} \tag{25}$$
$$\beta = \frac{\alpha}{1 - \alpha} \tag{26}$$
$$i_E = (\beta + 1) i_B \tag{27}$$

Early Effect and Output Resistance

$$i_C = I_S e^{v_{BE}/V_T} \left(1 + \frac{v_{CE}}{V_A}\right) \tag{28}$$
$$r_o = \frac{V_A}{I_C} \tag{29}$$

Small-Signal Parameters

$$V_T = \frac{kT}{q} \approx 25 \text{ mV (at room temperature)} \tag{30}$$
$$g_m = \frac{I_C}{V_T} \tag{31}$$
$$r_\pi = \frac{\beta}{g_m} = \frac{V_T}{I_B} \tag{32}$$
$$r_e = \frac{\alpha}{g_m} = \frac{V_T}{I_E} \tag{33}$$
$$r_\pi = (\beta + 1) r_e \tag{34}$$

Amplifier Configurations (Basic CE)

$$R_{in} = r_\pi \tag{35}$$
$$A_{vo} = -g_m R_C \tag{36}$$
$$G_v = -\left(\frac{r_\pi}{r_\pi + R_{sig}}\right) g_m (R_C \parallel R_L) \tag{37}$$

Amplifier Configurations (CE with Emitter Resistor R_e)

$$R_{in} = (\beta + 1)(r_e + R_e) \tag{38}$$
$$A_{vo} = -\frac{\alpha R_C}{r_e + R_e} \approx -\frac{g_m R_C}{1 + g_m R_e} \tag{39}$$